Tinker Tailor LLM Spy: Reconstructing "Deleted" Chats & Hijacking Sessions from Chromium LevelDB Caches

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I. Research Problem & Objective

Enterprise employees paste sensitive credentials and code blocks into AI portals, assuming deletion wipes local traces. In reality, browser LevelDB storage (IndexedDB) and IDE transcripts persist raw data on disk.Standard forensic suites fail to parse these V8 structures. We introduce Tinker Tailor LLM Spy to carve deleted AI logs and audit endpoints.

II. Technical Methodology & Carving Engines

- **V8 LevelDB Varint Carving:** Reverse-engineers Chrome SSTables, decoding varints to reconstruct ChatGPT/Gemini chat trees.
- **Claude TipTap Draft Recovery:** Extracts unsubmitted text buffers typed into Claude's prompt box before submission.
- **IDE Agent Carving:** Parses local JSONL transcript storage paths for Cursor and VS Code Copilot.

III. Technical Advantage Matrix vs. Legacy Forensic Suites

Feature	**Autopsy / Magnet AXIOM**	**Tinker Tailor LLM Spy**
ChatGPT V8 Tree Recovery	Generic Strings Only (Unstructured)	**100% Tree Recovery**
Claude Keystroke Drafts	0 Records Recognized	**Deduplicated Recovery**
Cursor & Copilot Parsing	Unsupported Schema	**Native Agent Crawler**
Secret DLP Alerts	Basic YARA Regex	**Real-time Alerts + Badges**
Execution Speed	Minutes to Hours	**51.9 milliseconds**

IV. Key Innovations & Mitigations

- **41ms Infostealer Vector:** A zero-dependency Python script (verity_stealer.py) harvests local LLM caches in 41 milliseconds.
- **DLP Engine & Takeover:** Web dashboard highlights leaked secrets and provides 1-click Playwright session hijacking takeover.
- **Mitigations:** Enforce client-side LevelDB encryption-at-rest and mandate log block zeroization upon session deletion.